

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Dekkshitha A (192572133) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Research Article Review

Selected Research Article

Title: Advancements in Deep Reinforcement Learning and Inverse Reinforcement Learning for Robotic Manipulation: Toward Trustworthy, Interpretable, and Explainable Artificial Intelligence

Authors: Mehmet Özalp, S. Uğur, and B. Gören

Journal: IEEE Access, Volume 12, pp. 51840–51858

Year: 2024

DOI: 10.1109/ACCESS.2024.3385426

Domain: Robotics / Artificial Intelligence / Robotic Manipulation

ML Topic: Deep Reinforcement Learning (DRL) / Inverse Reinforcement Learning (IRL) / Explainable Artificial Intelligence (XAI)

IEEE Xplore: <https://ieeexplore.ieee.org/document/10495698/>

DOI: <https://doi.org/10.1109/ACCESS.2024.3385426>

The article was published in IEEE Access, Volume 12, pages 51840–51858, in 2024.

Task 1: Article Summary

(a) Full Citation and Domain

- **Citation:** Recep Ozalp, Aysegul Ucar, and Cuneyt Guzelis, "Advancements in Deep Reinforcement Learning and Inverse Reinforcement Learning for Robotic Manipulation: Toward Trustworthy, Interpretable, and Explainable Artificial Intelligence," IEEE Access, vol. 12, pp. 51571–51614, 2024. DOI: 10.1109/ACCESS.2024.33854261.
- **Domain/Application Area:** Robotic Manipulation / Trustworthy, Interpretable, and Explainable Artificial Intelligence (XAI)²³.
- **Specific ML Problem:** Systematic taxonomic classification, literature review, and qualitative critique of Deep Reinforcement Learning (DRL) and Inverse Reinforcement Learning (IRL) methods applied to robotic manipulation over a five-year period (2018–2023)². The paper focuses on evaluating how these methods address problems such as perception, assembly, multitasking, transfer learning, human-robot interaction (HRI), and how they incorporate trust, interpretability, and explainability into black-box control policies².

(b) Summary and Objectives

The key objective of this article is to provide a comprehensive, structured literature review of the advancements in Deep Reinforcement Learning (DRL) and Inverse Reinforcement Learning (IRL) for robotic manipulation, with a particular focus on bridging the gap between high-performance control and the rigorous safety and auditability requirements of Trustworthy, Interpretable, and Explainable AI (XAI)²more_horiz.

The authors identified several critical research gaps in the literature: (1) traditional robotic control policies lack transparency, preventing human operators from understanding *why* a robot takes a specific action, which is a major barrier in high-stakes fields like medicine and smart manufacturing⁴more_horiz; (2) designing precise manual reward functions for complex, multi-stage manipulation tasks is notoriously difficult and prone to exploitation⁸⁹; and (3) both DRL and IRL face major robustness, safety, and scalability bottlenecks when transitioning from simulation to real-world environments¹⁰¹¹.

To fill these gaps, the authors propose a multi-dimensional taxonomic framework that maps five years of literature across task categories, summarizes the unique advantages and disadvantages of DRL vs. IRL, and details the emerging paradigms for injecting safety constraints, physical priors, and explainability metrics directly into the reinforcement learning loop²more_horiz.

Task 2: Methodology Analysis

(a) Machine Learning Techniques and Literature Pool Description

- **Taxonomic Categorization of ML Techniques:** Since this is a comprehensive literature review, the "methodology" represents the taxonomic categorization of RL and IRL techniques:
 1. **Model-Free DRL:** Categorized into Value-Based (e.g., Deep Deterministic Policy Gradient [DDPG], Twin Delayed DDPG [TD3], Soft Actor-Critic [SAC])¹³ and Policy-Based (e.g., Policy Gradient, Advantage Actor-Critic [A2C/A3C], Proximal Policy Optimization [PPO], Trust Region Policy Optimization [TRPO])¹⁴.
 2. **Model-Based DRL:** Categorized into Learning the Model (e.g., World Models, Imagination-Augmented Agents [I2A], Model-Based RL with Model-Free Fine-Tuning [MBMF])¹⁵ and Working on a Given Model (e.g., AlphaZero)¹⁵.
 3. **Inverse Reinforcement Learning (IRL):** Focuses on extracting reward/cost functions from expert demonstrations¹¹¹⁶, categorized into Maximum Entropy IRL, Bayesian IRL, and Adversarial IRL (e.g., GAIL)¹¹more_horiz.

4. **Trustworthy & Explainable AI (XAI) Techniques:** Integrates Shapley Additive Explanations (SHAP), formal verification, and digital twins to make learned policies auditable⁷more_horiz.

- **Literature Pool Analyzed:**

- **Source & Size:** The authors systematically extracted and analyzed literature published between 2018 and 2023²². The visual data in the paper shows they indexed hundreds of papers across Google Scholar and Web of Science²³. For example, the count of robotic manipulation DRL studies in Google Scholar alone exceeded 1,000 papers by 2023²³.
- **Curation & Preparation:** The authors structured their analysis into comparative matrices: Table 1 (listing 40+ prominent DRL-based manipulation articles)¹⁰, Table 2 (listing prominent IRL-based articles)¹¹, and Table 3 (listing articles focusing on trustworthy/interpretable/explainable AI in robotics)²⁴. These tables extract: (1) the specific paper citation, (2) the target task/problem, and (3) the proposed solution method¹⁰more_horiz.

(b) Mapping to CO 4 Course Topics

- **Value-Based and Policy-Based Reinforcement Learning:** This maps directly to the core RL syllabus. The paper details how model-free value-based methods (estimating expected cumulative reward $Q(s,a)$ via Bellman updates) and policy-based methods (directly optimizing policy parameters) are used for high-dimensional robotic control¹³14.
- **Inverse Reinforcement Learning (IRL) & Statistical Learning:** IRL maps to advanced statistical learning where the objective is to infer a latent variable—specifically, the hidden reward function $R(s,a)$ —from observed state-action trajectories generated by an expert policy, avoiding manual reward engineering¹¹16.
- **Trustworthy & Interpretable AI:** Maps to model interpretation and post-hoc explainability, demonstrating how complex non-linear policy networks can be evaluated using feature attribution methods (such as Shapley values)²⁰25.

Methodological Strengths & Limitations of the Review Approach

- **Strength — Bidirectional DRL/IRL Critique:** Unlike surveys that focus only on DRL or only on IRL, this paper provides a balanced, side-by-side comparison of both paradigms. In Table 4, they systematically contrast DRL's strength in overcoming analytical modeling of soft/deformable objects with its high sample-complexity limitations²⁶, directly against IRL's strength in learning without hand-crafted reward functions with its high sensitivity to expert noise and reward identifiability problems¹¹more_horiz.
- **Limitation — Lack of Quantitative Meta-Analysis:** The survey is purely qualitative and taxonomic. While it lists dozens of papers and their respective tasks, it does not provide a unified quantitative benchmark (e.g., comparing actual success rates, sample efficiency curves, or training hours of the reviewed algorithms on standardized robotics benchmarks like SoftGym or MuJoCo). This makes it difficult for a reader to objectively determine which specific algorithm is superior for a given manipulation task.

Task 3: Results & Critical Evaluation

(a) Key Performance Benchmarks & Growth Trends

The paper presents the growth trends and primary challenges identified across the reviewed literature. To illustrate the comparative advantages and disadvantages highlighted by the authors in Table 4, we can construct the following comparative matrix:

Comparative Summary of DRL and IRL in Robotic Manipulation:

The review contrasts the core capabilities and trade-offs of the two main paradigms:

Criterion / Paradigm	Deep Reinforcement Learning (DRL) ²⁶	Inverse Reinforcement Learning (IRL) ²⁶
Primary Advantages	- Performs exceptionally well on complex manipulation^{26,27} - Overcomes difficulties of analytically modeling soft/deformable objects^{26,28} - Handles full status observability in dynamic environments²⁶	- Learns optimal policies directly from human demonstrations^{11,26} - Eliminates the need to hand-craft complex reward functions^{11,26} - Easily captures latent human preferences^{26,29}
Primary Disadvantages	- Hyperparameter tuning (e.g., learning rates) is highly sensitive²⁶ - Extremely high sample complexity / trial-and-error costs^{8,26} - No inherent safety guarantees against dynamic obstacles^{10,26}	- High sensitivity to noisy or suboptimal expert demonstrations^{16,26} - Requires multiple representations to correctly understand rewards²⁶ - "Reward Identifiability" problem (multiple rewards explain one policy)¹¹more_horiz
Sim-to-Real Transfer	Sensitive to visual and dynamic gaps between simulation and reality ¹⁰	Highly complex; requires robust latent state representation ^{11,30}

Growth of Research Interest (2018 to 2023)²³:

- **DRL in Robotic Manipulation:** Exponential growth, with Google Scholar publications rising from ~80 in 2018 to over 1,000 in 2023²³.

- **IRL in Robotic Manipulation:** Gradual, steady growth, rising from ~30 publications in 2018 to nearly 200 in 2023²³.
- **Trustworthy/Explainable AI in Robotics:** Emerged dynamically, with publications jumping from 0 in 2018 to nearly 40 by 2023, signaling a major paradigm shift toward transparency²³.

(b) Critical Evaluation

- **Are the Reviewed Metrics Realistic:** The paper highlights that while many studies report high success rates in simulation environments (like Gazebo, Webots, or V-REP)²²³¹, these metrics are often highly unrealistic when applied to the real world. The "sim-to-real gap" (discrepancies in friction, contact physics, and sensor noise) acts as a massive barrier, meaning a simulated 99% success rate frequently drops significantly in physical deployment¹⁰.
- **Potential Biases in the Literature Pool:** There is a severe simulation bias in the indexed papers. Because real-world robot training is slow and risks damaging expensive hardware, the vast majority of reviewed RL/IRL methods are trained and tested primarily in idealized simulations, leaving physical safety constraints unproven.
- **Missing Experiments/Analysis in the Survey:** To strengthen this review, the authors should have included a cross-platform validation section. An analysis comparing how the same algorithm (e.g., SAC) performs across different physical robotic arms (e.g., Baxter vs. KUKA vs. Universal Robots) under varying lighting and wear-and-tear conditions would reveal real-world robustness.

(c) Real-World Applicability & Industry Scaling Challenges

- **Where to Deploy:** These techniques are actively being deployed in smart manufacturing assembly lines (Industry 5.0) for high-precision insertion tasks, automated warehouse sorting, surgical assistant robotics, and collaborative human-robot environments (HRC)^{6more_horiz}.
- **Challenges in Scaling to Industry:**
 1. **Safety Constraints:** Standard RL relies on unconstrained trial-and-error⁷³³. In a factory, a single erratic action by a heavy robot arm can destroy equipment or injure a human worker, making pure exploration non-viable without strict formal verification⁷¹⁰.
 2. **Data and Compute Inefficiency:** Training an agent to perform a complex bimanual manipulation task can require millions of interaction steps²⁶. Small-to-medium enterprises cannot afford the high compute budgets and lengthy training times.

3. **Ethical & Legal Black-Box Liability:** If an autonomous surgical robot or collaborative industrial gripper makes an error that causes harm, the lack of post-hoc explainability makes it legally impossible to trace whether the failure was due to sensor noise, policy misalignment, or an out-of-distribution state⁷⁹.

Task 4: Reflection & Learning Outcome

(a) **Three Key Insights Connected** This review deepened my understanding of the Reinforcement Learning syllabus (specifically Reward Function Design). I learned that hand-crafting scalar rewards is a critical bottleneck⁸. IRL elegantly bypasses this by treating the reward function as a latent variable to be statistically inferred from demonstrations, demonstrating a beautiful intersection of RL and Statistical Learning¹¹¹⁶.to CO 4 Syllabus

1. **The Intrinsic Reward Design Bottleneck:**
2. **Post-Hoc Explainability in Control Policies:** I realized that XAI is not just for computer vision or text classification. Applying feature-attribution techniques like Shapley Additive Explanations (SHAP) or formal methods to sequential decision-making policies allows us to understand which environmental states (e.g., joint angles, tactile pressures) most heavily influenced a robot's action selection²⁰²⁵.
3. **The Importance of Environmental Dynamics Modeling:** The clear distinction between model-free and model-based RL highlighted how a model acts as an inductive bias. Model-based architectures (like World Models) use statistical learning to predict environmental transitions $P(s'|s,a)$, allowing the agent to "imagine" trajectories and dramatically reduce real-world sample complexity¹⁴¹⁵.

(b) **Proposed Extension:** Explainable Multi-Task Imitation Learning via Causal SHAP

- **The Proposal:** I propose a framework that combines Multi-Task Inverse Reinforcement Learning (MT-IRL) with Causal Shapley Values to learn safe, multi-task robotic assembly policies while providing real-time, causal explanations of the robot's decisions¹⁸²⁵.
- **Algorithmic Enhancement:**
 - **IRL Layer:** Train a multi-task adversarial imitation learning policy using human expert demonstrations across diverse assembly tasks (e.g., peg-in-hole insertion, block stacking)¹⁸³².

- **Explainability Layer:** Integrate a post-hoc Causal SHAP pipeline²⁵. During execution, the model will output both the joint action and a real-time visual attention map indicating which sensory inputs (visual camera pixels, torque sensor readings) causally drove the action.
- **Safety Guardrail:** Implement scenario-based programming constraints that override the RL policy if the causal SHAP attribution of a critical safety feature drops below a threshold³⁴.
- **Feasibility & Expected Impact:** Highly feasible. The paper notes that both multitask IRL and post-hoc SHAP explanation tools exist independently but have not been formally combined into a unified, safe architecture¹⁸²⁵. This would drastically accelerate the trust and deployment of collaborative robots (cobots) in manufacturing, as human operators can immediately audit *why* a robot paused or adjusted its grip

